Convex-Chart Geometry of Nonlinear Staple Dispersion

21 July 2026

Abstract

We compare the Fréchet dispersion of an $SU(2)$ staple family with a pairwise gauge-invariant proxy and quantify the complete curvature loss on the certified local chart.

Let $S_1, \dots, S_m$ lie in a geodesic ball of radius $r < \pi/2$ about their Fréchet mean C, and put

$$D = \sum_i d(C, S_i)^2, \qquad P = \frac{1}{m} \sum_{i<j} d(S_i, S_j)^2, \qquad s_r = \frac{\sin r}{r}.$$

Theorem 1. *If the Karcher equation $\sum_i \log_C S_i = 0$ holds, then*

$$s_r^2 D \le P \le D.$$

Both quantities are invariant under a common endpoint gauge action.

Proof. Write $z_i = \log_C S_i$. Convexity of the ball and the differential singular values of the spherical exponential give $s_r \|z_i - z_j\| \le d(S_i, S_j) \le \|z_i - z_j\|$. The Euclidean identity $m^{-1} \sum_{i<j} \|z_i - z_j\|^2 = \sum_i \|z_i\|^2$ follows from $\sum_i z_i = 0$. Summing proves the claim. Bi-invariance proves gauge invariance. □

At the certified three-link staple radius $r = 0.01043241535$, the factor s_r^2 exceeds 0.9999, so convex-chart geometry causes negligible loss.

Remark 1. *This theorem is geometric. A normalized density comparison uniform over retained Yang–Mills boundary states is still required.*